



Serkan Sulun, PhD

Machine learning scientist

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PUBLICATIONS

- Sulun, S., & Viana, P.** (2026). Lightweight Speech Emotion Recognition with Disjoint Modality Fusion, INTERSPEECH 2026 Demo Session. (*accepted*)
- Sulun, S., Viana, P., & Davies, M. E. P.** (2026). Video Soundtrack Generation by Aligning Emotions and Temporal Boundaries. IEEE Transactions on Multimedia. (*in print*) [arXiv](#) | [code](#)
- Sulun, S.** (2025). Video-based Music Generation, PhD Thesis. [arXiv](#) | [code](#)
- Sulun, S., Viana, P., & Davies, M. E. P.** (2024). VEMOCLAP - A video emotion classification web application. 2024 IEEE International Symposium on Multimedia (ISM), 137–140. [DOI](#) | [arXiv](#) | [demo](#)
- Sulun, S., Viana, P., & Davies, M. E. P.** (2024). Movie trailer genre classification using multimodal pretrained features. Expert Systems with Applications, 258, 125209. [DOI](#) | [arXiv](#) | [code](#)
- Sulun, S., Oliveira, P., & Viana, P.** (2023). Emotion4MIDI: A Lyrics-based Emotion-Labeled Symbolic Music Dataset. EPIA 2023. Lecture Notes in Computer Science, vol 14116. Springer. [DOI](#) | [arXiv](#) | [code](#)
- Sulun, S., Davies, M. E. P., & Viana, P.** (2022). Symbolic Music Generation Conditioned on Continuous-Valued Emotions. IEEE Access, 10, 44617–44626. [DOI](#) | [arXiv](#) | [code](#)
- Sulun, S., & Tekalp, A. M.** (2021). Can learned frame prediction compete with block motion compensation for video coding? Signal, Image and Video Processing, 15(2), 401–410. [DOI](#) | [arXiv](#) | [code](#)
- Sulun, S., & Davies, M. E. P.** (2020). On filter generalization for music bandwidth extension using deep neural networks. IEEE Journal of Selected Topics in Signal Processing, 15(1), 132–142. [DOI](#) | [arXiv](#) | [code](#)
- Sulun, S.** (2018). Deep Learned Frame Prediction for Video Compression, Master's Thesis. [arXiv](#) | [code](#)

WORK EXPERIENCE

Computer Vision Center, Postdoctoral researcher	May 2026 – Present Barcelona, Spain
INESC TEC, Researcher	Mar. 2019 – Mar. 2026 Porto, Portugal
- Automatic video-based symbolic music generation - Finetuning vision language models for industrial safety	
University of Porto, Lecturer	Sep. 2023 – Present Porto, Portugal
Lecturer for the course Programming with Python	
Catolica Porto Business School, Lecturer	Mar. 2024 – Present Porto, Portugal
Lecturer for the course Python for Business Analytics	
Koc University, Research assistant	Sep. 2016 – Nov. 2018 Istanbul, Turkey
Video compression and frame prediction using deep neural networks	
Koc University, Teaching assistant	Sep. 2016 – Nov. 2018 Istanbul, Turkey
Preparing and leading laboratory sessions for the course, Programming with MATLAB	
Champalimaud Centre for the Unknown, Intern	Jul. 2013 – Oct. 2013 Lisbon, Portugal
- Processing neural signals and videos obtained by rat experiments - Designing the controller board. Article acknowledgment: DOI	
Sabanci University	Feb. 2010 – Jun. 2014 Istanbul, Turkey
Peer lecturer on freshman mathematics, physics, chemistry, and biology	

EDUCATION

University of Porto

PhD, Electrical and Computer Engineering

Present

Porto, Portugal

Koc University

MS, Electrical and Electronics Engineering (GPA: 3.81/4, converted to 19/20)

Nov. 2018

Istanbul, Turkey

Sabanci University

BS, Electronics Engineering (GPA: 3.86/4, converted to 20/20)

Jun. 2014

Istanbul, Turkey

MANAGEMENT AND LEADERSHIP

Principal Investigator: INESC TEC Seed project: Video and MIDI processing using deep neural networks.

Supervisor: Master's thesis: Emotion classification of MIDI lyrics using transformers

Supervisor: Bachelor's project: Speech emotion classification

Coordinator: Lab sessions and 5 assistants in the course, Introduction to MATLAB

SKILLS

Deep learning: Pytorch, Tensorflow, generative AI, transformers, adversarial learning, convolutional neural networks, vision language models, LoRA finetuning, cloud platforms (AWS, GCP, Azure), Slurm

Programming: Python, MATLAB, C++, Julia

Signal processing: MIDI, audio, image, video, natural language processing; video compression; computer vision

Miscellaneous: Machine learning, remote development, Git, LaTeX, Ubuntu, Bash

Music: Drumming, Cubase, Protools, Audacity, MIDI

INVITED TALKS

Introduction to deep learning, *Polytechnic Institute of Porto, Portugal*

Deep learning for music generation, *University of Porto, Portugal*

Deep learning for music information retrieval, *University of Porto, Portugal*

INTERVIEWS

SIC Noticias: Portugal vai receber Turquia

[Website](#) | [YouTube](#)

Observador: Generating music from videos

[Portuguese original](#) | [English translation](#)

Olhar No Progresso: Deep learning, multimedia, and music

[Website](#) | [YouTube](#)

LANGUAGES

English (C2), Portuguese (B1), Spanish (A1), Turkish (native)

ACTIVITIES

Finished in 04:20:05, in the 42 km. Porto Marathon

Drummer, in Santa Cecilia Music School rock band

Parachuting, in Turkish Aeronautical Association with 5-day training camp and 400 m. solo jump

Volunteer, Telfs Youth Center Austria (Erasmus+) and World Wide Opportunities on Organic Farms (WWOOF)